

Beatrice BATTISTI

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EDUCATION

Politecnico di Torino & Université de Bordeaux

Nov 2020 – Apr 2024

Doctor of Philosophy, Applied Mathematics and Mechanical Engineering Cotutelle programme, with high honours

Supervisors: Dr. Michel Bergmann, Prof. Giovanni Bracco

Defense: 22nd April 2024

Université Claude Bernard - Lyon 1 & Ecole Centrale de Lyon

Sep 2017 – Oct 2018

Master of Science, Fluid Mechanics and Energy

Ecole Centrale de Lyon

Sep 2016 – Oct 2018

Master of Science, General Engineering

Double Degree

Major in Aeronautics

Politecnico di Torino

Sep 2015 – Oct 2018

Master of Science, Aerospace Engineering and Astronautics

Double Degree, with high honours

Major in Aerodynamics

Politecnico di Torino

Sep 2012 – Sep 2015

Bachelor of Science, Aerospace Engineering

with high honours

RESEARCH INTERESTS

I am interested in developing numerical schemes for partial differential equations to describe the physics of environmental phenomena. My focus is on the numerical modeling of multi-phase flows and the implementation of model order reduction methods for non-linear problems.

CURRENT POSITION

CNRS, Université Grenoble Alpes

Oct 2026 – present

Junior Researcher (Chargée de Recherche)

Laboratoire Jean Kuntzmann (LJK)

PROFESSIONAL AND RESEARCH EXPERIENCE

CNRS, Université Savoie Mont Blanc (LAMA)

Chambéry, France

Post-doctoral fellowship - Supervisors: Walter Boscheri, Alain Burgisser

Sep 2025 – Aug 2026

- ERUPTA: Efficient and Reliable numerical methods to Unravel Processes in magma Transport at Active volcanoes

CNRS, Université Savoie Mont Blanc (LAMA)

Chambéry, France

Post-doctoral fellowship - Supervisor: Walter Boscheri

Sep 2024 – Aug 2025

- Numerical simulation and analysis of multiphase flows applied to volcanic phenomena

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After PhD

Before PhD

Politecnico di Torino

Turin, Italy

Pre-Doctoral program

Apr 2020 – Oct 2020

- Wave energy converter array development

Politecnico di Torino

Turin, Italy

Research scholarship

Jan 2020 – Apr 2020

- Development of a CFD in-house code - Application to the wave energy

IKOS Consulting, at ALSTOM

La Rochelle, France

Engineering full-time job

Mar 2019 – Dec 2019

- Numerical simulation of train dynamics in extreme conditions - Application to trams and to trains for the "TGV 2020" project

INRIA Bordeaux - Sud-Ouest

Internship

- Numerical simulation of undulating bores in open channels and study of the influence of the sloping banks on the flow dynamics

Naval Group

Internship

- Hydrodynamics of the towed "V-Wing" systems and dimensioning of the drop cable

Bordeaux, France

Apr 2018 – Sep 2018

Ollioules, France

Apr 2017 – Aug 2017

RESEARCH STAYS

Optimad Srl

Secondment, ARIA (Accurate ROMs for Industrial Applications) project

- Body-less approach for the numerical modeling of wave energy converters

Ghent University

WECANet Short Term Scientific Mission (STSM)

- Numerical modeling of the far-field effects of a PeWEC farm

CIRM

Research project in the context of the summer school CEMRACS 2021

- Model Order Reduction of one-dimensional non-linear transport PDEs in porous media

Turin, Italy

Winter 2023 – 3 months

Ghent, Belgium

Fall 2022 – 3 weeks

Luminy, France

Summer 2021 – 5 weeks

AWARDS, DISTINCTIONS AND RESEARCH GRANTS

- MSCA Seal of Excellence, Marie Skłodowska-Curie Actions Postdoctoral Fellowships (European Commission). Project: *LIMEN: Leading Innovation in Multiphase Environmental Numerics*. Final score: 89.60/100. Winter 2026.
- WECANet COST Action Short-Term Scientific Mission (STSM) Fellowship, Ghent University - Numerical modeling of WEC farms. 4,000€. Fall 2022.
- CEMRACS Summer School Fellowship, CIRM - Reduced-order modeling of transport PDEs in porous media. *Full support*. Summer 2021.

ORGANIZATION OF SCIENTIFIC MEETINGS

Université Savoie Mont Blanc

Co-organizer of the Girls and Maths day

- with Laure Bastide, Georges Comte and Céline Labart (LAMA Equality Committee)

LAMA - Université Savoie Mont Blanc

Organizer of the PhD and Postdoc days in Mathematics of USMB

Université Savoie Mont Blanc

Co-organizer of the 3C conference

- 3C: Challenges in Computational methods for Complex environmental applications
- with Laure Bastide, Walter Boscheri and Arnaud Duran

LAMA - Université Savoie Mont Blanc

Co-organizer of the PhD and Postdoc day in Mathematics of USMB

- with Cassandre Lebot

Chambéry, France

May 2026

Yenne, France

October 2025

Chambéry, France

May 2025

Chambéry, France

November 2024

PUBLICATIONS

Summary (as of October 2026):

Google Scholar: 17 publications, 91 citations, h -index = 5.

Scopus: 10 publications, 39 citations, h -index = 3.

The complete list of publications is provided in Annex 1.

Dissertations

- Multi-fidelity multi-scale numerical modeling of wave energy converter farms. Doctoral thesis, 2024.
- Modeling of the upstream propagation of coastal surface waves into open channels - Application to the dynamics of undulating bores. Master thesis, 2018.
- Implementation of an auxiliary instrument for the design of a wingsail. Bachelor thesis, 2015.

TEACHING ACTIVITIES

2026/2027 – 1st semester (10 hours)

Université Grenoble Alpes, Ensimag
Course "PDEs and numerical methods" (TP)

Grenoble, France
~ 20 students

2025/2026 – 1st semester (45 hours)

Université Savoie Mont Blanc
Course "Partial differential equations and discretization" (TP)
Course "Mathematical modeling and scientific computing" (TP)
Course "Numerical analysis" (TD)
Course "Discrete probability" (TD)

Chambéry, France
~ 15 students
~ 15 students
< 10 students
~ 25 students

2023/2024 – 1st semester (48 hours)

Enseirb-Matmeca
Course "Algorithm and Programming in Fortran 90" (TP)

Bordeaux, France
~ 15 students

2022/2023 – 2nd semester (20 hours)

Enseirb-Matmeca
Course "Algorithm and Programming in Fortran 90" (TP)

Bordeaux, France
~ 15 students

PRESENTATIONS

Invited Talks Typically ≥ 30 min presentation; full support provided (conference fee and accommodation covered).

MACS Seminar at IMAG. Montpellier, France - Mar 2026.
AMAC Seminar at LJK. Saint-Martin-d'Hères, France - Feb 2026.
ETNA Conference. Catania, Italy - Nov 2025.
JEARA Conference. Lyon, France - Nov 2025.
EDP Seminar at LAMA. Le Bourget-du-Lac, France - Sep 2025.
INORE Symposium. Zarautz, Spain - Oct 2022.

Contributed Talks

Model reduction and CFD: a workshop to celebrate the contributions of the actual AI. Bilbao, Spain - Oct 2026.
Digital Ocean for the Future, Inria Webinar Series. Online - Jan 2026.
ARIA Workshop. Bidart, France - Sep 2025.
Mathematics of Compressible Fluids Workshop. Clausthal, Germany - Jun 2025.
Shark-FV Conference. Porto, Portugal - Jun 2025.
3C Conference. Chambéry, France - May 2025.
EWTEC Conference. Bilbao, Spain - Sep 2023.
ECCOMAS MARINE Conference. Madrid, Spain - Jun 2023.
ECCOMAS COUPLED PROBLEMS Conference. Chania, Crete, Greece - Jun 2023.
SIAM CSE Conference. Amsterdam, Netherlands - Mar 2023.
RENEW Conference. Lisbon, Portugal - Nov 2022.
Closure of STSM Project at Ghent University. Gent, Belgium - Sep 2022.
HYWEC Workshop. Bilbao, Spain - Jun 2022.
7th Wave Energy Workshop. Turin, Italy - Apr 2022.
Closure of CEMRACS 2021 Research Session at CIRM. Luminy, France - Aug 2021.

POSTERS

Numerical Aspects of Hyperbolic Balance Laws and Related Problems – Young Researchers Conference, Ferrara, Italy - Dec 2024.
Workshop on "Reduced-order models at work: Industry and Medicine", Bordeaux, France - Mar 2023.
4th WECA Net Assembly, Ghent, Belgium - Mar 2023.
INORE Symposium 2022, Zarautz, Spain - Oct 2022.
Workshop on "Reduced-order models at work: Industry and Medicine", Bordeaux, France - Mar 2022.

OUTREACH

- Presentation on "Stereotypes in Science" for high school girls during the "Filles, maths et informatique : une équation lumineuse" Day, Chambéry, France - May 2026.
- Simulating complex real-world phenomena: can we make it affordable? Scientific presentation at the EDMi Doctoral School Day in Mathematics and Computer Science, Bordeaux, France, Apr 2026.
- Participation in the mathematics booth for the science festival "Fête de la Science", Chambéry, France - Oct 2025.
- Participation to the speed meeting with high school young women during the "Filles, maths et informatique : une équation lumineuse" day, Chambéry, France - May 2025.
- Vague océanique : espoir d'une énergie renouvelable ou crainte de la montée des eaux ? Amphi Pour Tous Conference, Chambéry, France - Oct 2024.
- Numerical modeling of wave energy converter farms. Seminar for the Lambda Team (association of the Bordeaux PhD students in Mathematics), Bordeaux, France - May 2023.
- The path to the PhD and an introduction to the PhD project, Talk with Master's students at ENSEIRB-MATMECA, Bordeaux, France, Nov 2022.

LANGUAGES

- **Italian** Native language.
- **French** Fluent. DALF C2 (2018).
- **English** Fluent. TOEFL iBT : 100 (2018). GRE General Test : 156-160-3.5 (2018). TOEIC (2017).
- **Spanish** Basic communication skills.

PROGRAMMING SKILLS

Advanced

Fortran
Matlab
Python

Intermediate

Java
Julia

Basic

C/C++
HTML/CSS

INTERESTS



Environment



Climbing



Mountains



Paragliding



Reading



Cycling

Annex 1.

PUBLICATIONS

Referred Publications in International Journals

- Battisti B., Carlino M. G., Boscheri W., Iollo A. (2026). *Implicit-explicit collocated Reduced Order Model schemes for multiscale systems of PDEs*. preprint. PDF
- Battisti B., Boscheri W. (2025). *A linearly implicit shock capturing scheme for compressible two-phase flows at all Mach numbers*. Journal of Computational Physics (JCP), 539:114227. DOI
- Battisti B., Bracco G., Bergmann M. (2024). *A multi-fidelity model for wave energy converters*. International Journal for Numerical Methods in Fluids (IJNMF), 97:427-445. DOI
- Battisti B., Giorgi G., Verao Fernandez G. (2024). *Balancing power production and coastal protection: A bi-objective analysis of Wave Energy Converters*. Renewable Energy, 220:119702. DOI
- Cervelli G., Battisti B., Mattiazzo G. (2022). *On the influence of multidirectional irregular waves on the PeWEC device*. Frontiers in Energy Research, 20:908529. DOI

Conference Proceedings Publications

- Boscheri W., Battisti B., *A penalized IMEX finite volume scheme for the isentropic Euler system at all Mach numbers*. Proceedings of the HYP 2024 conference, accepted, 2025. PDF
- Battisti B., Giorgi G., Verao Fernandez G. (2024). *Enhancing synergy for power generation and coastal protection through wave energy converters*. Innovations in Renewable Energies Offshore - RENEW, Lisbon. PDF
- Battisti B., Giorgi G., Verao Fernandez G., Troch P. (2023). *Multi-query analysis of a PeWEC farm*. Proceedings of the European Wave and Tidal Energy Conference - EWTEC, vol. 15, Bilbao. PDF
- Battisti B., Blickhan T., Enchery G., Ehrlacher V., Lombardi D., Mula O. (2023). *Wasserstein model reduction approach for parametrized flow problems in porous media*. ESAIM: Proceedings and Surveys, vol. 73, p. 28-47, Luminy. DOI
- Battisti B., Bracco G., Bergmann M. (2022). *Multi-fidelity modeling of wave energy converter farms*. Trends in Renewable Energies Offshore - RENEW, p. 351-357, Lisbon. website
- Niosi F., Battisti B., Sirigu S.A. (2022). *Influence of hydrodynamic interactions on the productivity of PeWEC wave energy converter array*. International Conference on Electrical, Computer, Communications and Mechatronics Engineering - ICECCME, Maldives. PDF
- Casalone P., Dell'Edera O., Fontana M., Battisti B., Mattiazzo G. (2022). *Solutions to Wave Damping Over Time in CFD RANS Simulations Due to Exponential Generation of Numerical Turbulence*. Proceedings of the International Conference on Offshore Mechanics and Arctic Engineering - OMAE, vol. 8, Hamburg. website
- Fontana M., Casalone P., Dell'Edera O., Niosi F., Battisti B., Mattiazzo G. (2021). *Viscous damping analysis of WEC's hull in yaw motion - Methodology and viscous damping parameters identification*. Proceedings of the European Wave and Tidal Energy Conference - EWTEC, Plymouth. website